

Yufeng Gu

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EDUCATION

University of Michigan

Ph.D. Student, Computer Science and Engineering Department

Advisor: Prof. Reetuparna Das, GPA: 3.88/4.00

Ann Arbor, MI, USA

Sep. 2020 - Apr. 2026

Zhejiang University (ZJU)

B.Eng., College of Information Science and Electronic Engineering

GPA: 3.96/4.00

Hangzhou, China

Sep. 2016 - June 2020

PROFESIONAL EXPERIENCE

University of Michigan

Graduate Research Assistant, Advisor: Prof. Reetuparna Das

Optimized systems and architectures for emerging applications, including Generative AI and genome sequencing analysis. Proposed solutions across domain specific accelerator, near-memory processing and QoS-aware hardware resource scheduler.

Ann Arbor, MI, USA

Sep. 2020 - Aug. 2025

Tenstorrent Inc.

Performance Architect Intern, Manager: Wei-han Lien

Built performance model for RISC-V CPU and AI accelerator.

Santa Clara, CA, USA

May 2023 - Aug. 2023

Intel Labs

Graduate Research Intern, Manager: Nilesh Jain

Optimized AI QoS on next generation heterogenous datacenter platform.

Hillsboro, OR, USA

June 2022 - Aug. 2022

Yale University

Undergraduate Research Intern, Advisor: Prof. James Duncan

Interpretation on ASD with fMRI and Deep Learning Models.

New Heaven, CT, USA

Nov. 2019 - Mar. 2020

École Polytechnique Fédérale De Lausanne (EPFL)

Undergraduate Research Intern, Advisor: Prof. Babak Falsafi

Patched QFlex Simulator and accelerated activation functions for DNN.

Lausanne, Switzerland

July 2019 - Sep. 2019

PEER-REVIEWED CONFERENCE/JOURNAL PUBLICATIONS

* equal contribution

Yufeng Gu*, Alireza Khadem*, Sumanth Umesh, Ning Liang, Xavier Servot, Onur Mutlu, Ravishankar Iyer, Reetuparna Das. "PIM is All You Need: A CXL-Enabled GPU-Free System for LLM Inference." In Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25).

Alireza Khadem, Daichi Fujiki, Hilbert Chen, **Yufeng Gu**, Nishil Talati, Scott Mahlke, Reetuparna Das. "Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing." In 2025 IEEE International Symposium on High-Performance Computer Architecture (HPCA'25). **Distinguished Artifact Honorable Mention**

Yufeng Gu, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." In Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA'23). **Communication of ACM (CACM) Research Highlights**

Arun Subramaniyan, **Yufeng Gu**, Timothy Dunn, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, and Reetuparna Das. "GenomicsBench: A Benchmark Suite for Genomics." In IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS'21).

Xiaoxiao Li, **Yufeng Gu**, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. "Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results." Medical Image Analysis 65: 101765. *IF = 11.1*

Xiaoxiao Li, Yuan Zhou, Nicha C. Dvornek, **Yufeng Gu**, Pamela Ventola, and James S. Duncan. "Efficient Shapley Explanation for Features Importance Estimation Under Uncertainty." In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI'20).

UNDER REVIEW CONFERENCE SUBMISSIONS

Yufeng Gu, Vui Seng Chua, Nilesh Jain, Ravishankar Iyer, Reetuparna Das. "Farm: Fast Resource Management for QoS-Aware Co-Location of ML Inference." *Under submission*

Sumanth Umesh, Ning Liang, **Yufeng Gu**, Reetuparna Das. "Addressing In-Memory Database Bottlenecks with CXL Memory Expansion." *Under submission*

WORKSHOP PUBLICATIONS

Noah Kaplan, **Yufeng Gu**, Reetuparna Das. "Understanding and Characterization of Pangenomics." in International Workshop on Domain Specific System Architecture (DOSSA) co-located with ISCA 2024.

AWARDS & HONORS

Distinguished Artifact Honorable Mention (HPCA 2025) *March 2025*

HPCA Travel Grant *March 2025*

Communication of ACM (CACM) Research Highlights *July 2024*

GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis. *CACM Research Highlights section selects 24 paper from all ACM conferences per year, reprinting the most significant and influential results across Computer Science.*

University of Michigan Rackham Graduate Student Research Grant (\$3000) *Apr. 2024*

ISCA Travel Grant *July 2023*

University of Michigan Rackham Travel Grant *2023, 2025*

Fellowship of Summer@EPFL (2% applicants awarded) *July 2019*

Tang Lixin Fellowship (2% students awarded) *Nov. 2017, 2018, 2019*

Outstanding Student Leaders in Zhejiang University (3% students awarded) *Oct. 2017, 2019*

First-Class Scholarship for Outstanding Students (2% students awarded) *Oct. 2017*

TECHNICAL SKILLS

Programming: C/C++, CUDA, Python, Verilog, Bash

Tools: Linux, Docker, Git, Slurm, PyTorch, Synopsis Design Compiler

SERVICES

Artifact evaluation reviewer for IISWC 2021, ISCA 2023, 2024.

Reviewer for University of Michigan graduate student admission

Jan. 2021, 2022

Voluntary teacher in Tuanlin Primary School, Guizhou, China

Aug. 2017